



10 (a) State Hubble's law.

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(b) A star in a distant galaxy emits radiation that has a maximum intensity of emission at a wavelength of $4.62 \times 10^{-7} \text{ m}$.

Observations of the galaxy made on the Earth detect the maximum intensity of emission from the star at a wavelength of $4.91 \times 10^{-7} \text{ m}$.

(i) Explain why the observed wavelength and the emitted wavelength have different values.

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(ii) Calculate the speed of the star relative to the Earth.

speed = m s^{-1} [2]

(iii) The wavelength of maximum intensity of emission is used to determine a value for the surface temperature of the star.

Explain how the temperature determined using the observed wavelength compares with the true value of temperature determined using the emitted wavelength.

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